

CONCEPT MAP 4: VIRTUALIZATION AND WEB SERVICES AS ENABLERS OF CLOUD COMPUTING

Project Title : Architectural Analysis and Deployment of Serverless Computing in Modern Cloud Platforms

VIRTUALIZATION & WEB SERVICES

1. VIRTUALIZATION

- Hypervisor
- Virtual Machines
- Resource Sharing
- Hardware Abstraction

2. CONTAINERIZATION

- Containers
- Docker
- Kubernetes
- Lightweight Deployment

3. WEB SERVICES

- REST APIs
- SOAP
- HTTP/HTTPS
- Data Exchange

4. CLOUD INTEGRATION

- API Gateway
- Microservices
- Event-Driven Architecture
- Serverless Integration

5. VIRTUALIZATION BENEFITS

- Resource Utilization
- Scalability
- Flexibility
- Cost Reduction
- Isolation

6. WEB SERVICE BENEFITS

- Interoperability
- Communication
- Automation
- Integration
- Accessibility

7. CONNECTION TO THE PROJECT

- Enables Serverless Computing
- Supports Cloud Applications
- Improves Resource Management
- Foundation for Modern Cloud Platforms

SERVERLESS COMPUTING

Architectural Analysis and Deployment of Serverless Computing in Modern Cloud Platforms